BLAKE J. GISCLAIR

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APPLIED MATHEMATICS Ph.D. STUDENT

MACHINE LEARNING | AEROSPACE RESEARCH

EDUCATION

Florida Institute of Technology, Melbourne, FL
Ph.D. in Applied Mathematics **2026-Present**
B.S. in Applied Mathematics, *magna cum laude* **2026**
Undergraduate GPA: 3.9 | **Apollo Honors**

RESEARCH & AEROSPACE EXPERIENCE

Neural Transmissions (NETS) Lab - Data Scientist **Fall 2023-Present**

- Work toward developing machine learning (ML) algorithms to automatically detect nuclear explosions for the Air Force Technical Applications Center (AFTAC).
- Work with geophysical signals processing using machine learning (ML) to categorize anomalous signals.
- Develop entropy based ensembling techniques for the US Army Engineer Research and Development Center (ERDC).
- Work on National Science Foundation (NSF) funded research analyzing Hyperspectral Imaging (HSI) data to detect and analyze cancer cells.
- Work as co-author on papers in the field of ML explainability techniques.
- Research efficacy of edge deployment for Visual Transformer (ViT) based models on low size, weight, and power (SWaP) satellites into space as an alternative to traditional Convolutional Neural Networks (CNNs).
- Presented research findings at the Florida Mathematical Association of America (FL-MAA) Conference - Spring 2023. Awarded best undergraduate presentation.
- Completed mathematical research deriving closed expressions for analyzing the crossing patterns of stochastic networks.

NASA Goddard Space Flight Center (GSFC) - Machine Learning Intern **May-August 2025**

Magnetospheric Multiscale (MMS) Mission

- Worked in the Ground System Software branch of GSFC supporting the Science Data Processing branch and in conjunction with members of the Heliophysics Science Division.
- Aided in telemetry analysis to support the Magnetospheric Multiscale Mission (MMS).
- Learned about, utilized, and implemented Temporal Convolutional Networks (TCNs) to detect anomalous behavior in MMS satellites with the goal of predicting and preventing critical failures.

Instructor at SPARK Innovative Laboratories **Spring 2026**

- Taught algebra, chemistry, and forensics to home school students, grades 6-12, in the Melbourne, FL area.
- Planned and executed weekly lesson plans to engage and educate students.

CONFERENCE PRESENTATION

Meni, M. J., Gisclair, B., White, R. T., & Mahendrakar, T. (2025, August 13). *PEEK-guided neural network pruning for deployment on low SWaP hardware*. Small Satellite Conference, Salt Lake City, UT.

MANUSCRIPT UNDER REVIEW

Meni, M. J., Gisclair, B., Niwas, M., & White, R. T. (2026). *PEEK variance: An information-theoretic metric unifying interpretability, optimization, and efficiency in deep neural networks*. Manuscript under review.

TECHNICAL SKILLS

Programming & Scientific Computing: Python, C++, Java, MATLAB, NumPy, SciPy, pandas
Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, CNNs, multimodal models, anomaly detection, model explainability
Computing & Development: Git/GitHub, Linux, CUDA, Jupyter
Mathematics & Methods: ODEs/PDEs, numerical methods, statistical/stochastic analysis, signal processing, ensemble methods

HONORS & AWARDS

Astronaut Scholarship Foundation Candidate | **Lockheed Martin STEM Scholarship** | **Dr. Vangipuram Lakshmikantham Math Scholar Fellowship** | **Phi Eta Sigma Honor Society** | Florida Tech Robotics Scholarship | Holt Family Foundation Scholarship | Panther Fund Scholarship